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Human Cognitive Benchmarks Reveal Foundational Visual Gaps in MLLMs

VisFactor: 23 Frontier Models Score Below 31% on Psychometric Visual Tests Normed to Human Populations

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Multimodal LLMs score impressively on academic visual benchmarks, yet those benchmarks allow language priors and common-sense shortcuts to compensate for weak raw visual perception. VisFactor digitizes 20 vision-centric subtests from the Factor-Referenced Cognitive Tests (FRCT)—a decades-old psychometric battery designed to isolate specific human visual faculties—and reveals that 23 frontier MLLMs, including GPT, Gemini, and Qwen families, achieve a best score of 30.17 out of 100. The human norm is approximately 80. Neither scale nor chain-of-thought prompting closes the gap, exposing foundational deficits in spatial reasoning and visualization that conventional benchmarks conceal.

AT A GLANCE

Problem: Academic MLLM benchmarks allow language shortcuts and miss the low-level visual cognition that human vision relies on.

Why it matters: Systems deployed in visually demanding roles (medical imaging, robotics, design) require the cognitive visual capabilities that current benchmarks do not probe.

Method: VisFactor digitizes 20 FRCT vision subtests spanning spatial reasoning, perceptual speed, visualization, and visual memory into zero-shot multiple-choice questions.

Result: Best of 23 frontier MLLMs scores 30.17/100; human norm $\approx$ 80; no model reaches the human level on any domain.

Evidence: 23 MLLMs (GPT, Gemini, Claude, LLaMA, Qwen, InternVL); ablations confirm visual engagement, rule out position bias.

The Big Picture

The standard narrative in multimodal AI is one of rapid convergence: each new MLLM generation narrows the gap to human performance on VQA, image captioning, and visual reasoning benchmarks. This narrative is misleading because those benchmarks are designed by ML researchers and systematically admit language-based strategies that bypass genuine visual computation.

Cognitive psychologists have known for decades that human visual cognition is a multi-factor system encompassing spatial visualization, perceptual speed, form completion, and short-term visual memory—each measurable independently with psychometric instruments normed to large human populations. VisFactor applies those instruments to MLLMs and finds a strikingly different picture: all frontier models score far below the human norm on every domain, with the largest gaps in 3D-to-2D spatial transformations and form completion.

Why Existing Benchmarks Fall Short

Language shortcuts: Many VQA questions can be answered partially from the question text alone; models with strong language priors achieve inflated scores without parsing the image carefully.

Coarse granularity: Aggregate accuracy on tasks like MMBench or MMMU merges many heterogeneous subtasks, making it impossible to identify which specific visual process is failing.

Saturation and ceiling effects: Frontier models now saturate several prominent benchmarks, leaving no headroom to distinguish capability levels among top systems.

Lack of psychometric grounding: Existing benchmarks carry no normative calibration against human populations, so "state-of-the-art" accuracy has no reference point against actual human visual cognitive ability.

Core Mechanism: VisFactor

VisFactor digitizes 20 vision-centric subtests from the FRCT battery, grouped into four domains:

Spatial Reasoning: Card Rotations, Paper Folding—tests of mental rotation and spatial visualization in 3D.

Perceptual Speed: Finding A's, Number Comparison—tests of rapid figure-ground segregation and symbol matching.

Visualization: Surface Development, Form Board—tests of 3D-to-2D reasoning and shape completion.

Visual Memory: Shape Memory, Position Memory—tests of short-term retention of visual configurations.

Each item is digitized as a multiple-choice question. The FRCT scoring key provides ground-truth answers. No image captioning or text-only reasoning is possible on any item; answering requires genuine visual computation.

Technical Formulation

Let M be an MLLM, $\mathcal{Q} = \{(q_i, A_i, a_i^*)\}_{i=1}^N$ the question set (image q_i, options A_i, correct answer a_i^*). The VisFactor score is:

$$\text{Score}(M) = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[M(q_i, A_i) = a_i^*] \times 100$$

Domain subscores partition the sum over the four cognitive domains. Random baseline performance is 25 (four options). Human FRCT norm is approximately 80. The gap $\Delta(M) = 80 - \text{Score}(M)$ measures how far a model is from average human performance.

Learning or Inference Procedure

VisFactor is a zero-shot evaluation protocol:

1. Present the model with the item image and the four multiple-choice options.
2. Use a fixed instruction: "Which option is correct? Answer with a single letter."
3. Record the model's response; compare to the FRCT scoring key.
4. Compute domain subscores and total score.

A chain-of-thought variant instructs models to explain their reasoning before answering. Both standard and CoT protocols are reported.

What the Guarantee Says

VisFactor's psychometric grounding provides several validity properties:

- **Construct validity:** FRCT subtests were validated against factor-analytic studies of human visual cognition; each subtest measures a theoretically distinct visual factor.
- **Convergent validity:** High within-domain correlations across models confirm that subtests in the same domain measure a common latent visual ability in MLLMs.
- **No language-shortcut confound:** Removing the image reduces all models to chance (25%), confirming that scores reflect visual engagement rather than language priors.

Experimental Findings

- **Best model:** Scores 30.17/100; human norm ≈ 80. All 23 models score between 25 (chance) and 31.
- **Domain asymmetry:** Perceptual speed $\approx 45\%$; spatial reasoning $\approx 28\%$; visualization $\approx 20\%$; visual memory $\approx 22\%$.
- **Scale does not help:** Larger models within each family show no consistent improvement on VisFactor, unlike most academic visual benchmarks.
- **Chain-of-thought:** Marginal improvement on spatial reasoning; slight degradation on perceptual speed (CoT slows the rapid comparison process the subtest is designed to measure).
- **No saturation:** No model approaches the human norm on any domain, leaving substantial room for future progress.

Ablations and Interpretation

Image removal: All models score $\approx 25\%$ when the image is removed and only the question text and options are provided, confirming genuine visual engagement.

Option order shuffling: Results are stable under option permutation (variance < 2 percentage points), ruling out position bias as a confound.

Item difficulty calibration: Items requiring mental rotation of complex 3D figures are the hardest; items requiring simple symbol comparison are easiest—matching the psychometric structure designed for humans. The cross-model difficulty ordering correlates with human item difficulty at $r = 0.73$.

Closed-source vs. open-source models: Closed-source models (GPT, Gemini, Claude) score marginally higher than open-source models of comparable size, but the gap is small relative to the distance from the human norm.

Reference

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